CH1 Axioms of Probability

1.2 Sample space and events

Definition

Experiment: 一個機率問題是由一個機率事件 (e.g. tossing a die)

Outcome: 產生結果 (A.K.A sample point)

Sample space: 所有可能的結果集合 (e.g. 丟硬幣 {Head, Tail})

Event: 所有可性的子集合 (e.g. 硬幣頭在上的可能性 {Head})

Ex 1.3 (An infinite sample space)

Measuring the lifetime of a light bulb, where last at least 100 hrs

$S=\{x:x\geq 0\}\quad E=\{x:x\geq 100\}$

Ex 1.4 (An finite sample space) (a.k.a Classical Probability)

All families with 1, 2, or 3 children (genders specified)

$S=\{b,g,bg,gb,bb,gg,bbb,bgb,bbg,bgg,ggg,gbg,ggb,gbb\}$

Definition

Occur: Event E has occured if result contains a element in E

$E\subset F$	E is a subset of F
$E\cap F=EF$	Intersection of E and F
$E\cup F$	Union of E and F
$E-F$	The set difference of E and F
$E^c=\bar{E}$	All element excluding E = $S-E$
$\cup_{i=1}^{n}E_i$	$E_1\cup E_2\cup E_3...E_n$
$\cap_{i=0}^{n}E_i$	$E_1\cap E_2\cap E_3\cap...\cap E_n$
$,\cup_{i=1}^{\infty}E_i,\cap_{i=1}^{\infty}E_i$	

Sample space and events

Commutative laws (交換律) EUF=FUE , EF=FE

Associative laws (結合律) EU(FUG)=(EUF)UG

Distributive laws (分配律) $(E\cup F)H=(EH)\cup(FH)$

$(EF)U\cup H=(E\cup H)(F\cup H)$

De Morgan's law 1st rule $(E\cup F)^c=E^cF^c$

2nd rule $(EF)^c=E^c\cup F^c$

E and F are mutually exclusive if they have nothing in common$EF=\phi$

無中生有 $E=ES=E(F\cup F^c)=EF\cup EF^c$ (也是因為 EF 與 EF^c 是mutually exclusive)

1.3 Axioms of probability

S (sample space): 所有的可能性的集合

A (event): 其中可能性的集合

Pr (Function): 發生 A 事件的機率為 P(A)

Axioms 1: P(A) ≥ 0 機率必定大於 0

Axioms 2: P(S) ≤ 1 機率最大為 1

Axioms 3: 只有在所有 event 皆 mutually exclusive 時，機率才為每個 event 的和

If $\{A_1, A_2, A_3, ..., A_n\}$ are mutually exclusive then: $P(\cup_{i=0}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

Classical definition of Probability

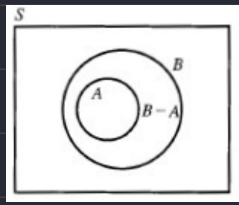
1. Sample space is finite
2. Each event is equally likely to occur (shares a same probability)

$P(A) = N(A)/N(S)$ ... N(A) is the number of sample points of A.

1.4 Basic Theorem

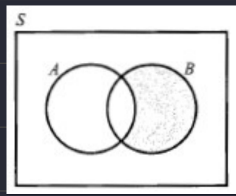
Theorem 1.4: $P(A^c) = 1 - P(A)$

Theorem 1.5: If $A \subset B$, then $P(B - A) = P(BA^c) = P(B) - P(A)$



Corollary If $A \subset B$, then P(A) ≤ P(B)

Theorem 1.6: $P(A \cup B) = P(A) + P(B) - P(AB)$



- **Inclusion-Exclusion Principle**

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = \sum P(A_i) - \sum P(A_i A_j)$$
$$+ \sum P(A_i A_j A_k) - \ldots + (-1)^{n-1} P(A_1 A_2 \ldots A_n)$$

1.5 Continuity of probability function (Omission)

- **Recall the continuity of a function f: R→R**

$$\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n)$$

for every convergent seq $\{x_n\}$ in R.

- **The continuity of probability function is similar.**
- **Def. A seq $\{E_n, n>=1\}$ of event of a sample space is called increasing if**

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots E_n \subseteq E_{n+1} \subseteq \cdots;$$

it is called decreasing if

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots E_n \supseteq E_{n+1} \supseteq \cdots.$$

1.6 Probabilities 0 and 1